CHAPTER 5

Trigonometric functions

S75, S76, S78, S79

Before you begin · Unit circle

Without a calculator, in six minutes. Answer briefly. Use radians, signs and differentiation. Then choose the targeted review indicated under the questions.

1. Convert 180 $^\circ$ to radians.
2. Give sin 0 and cos 0.
3. Give the sign of sin x on $]0;\pi[$.
4. Solve cos $x = 0$ on $[0;2\pi]$.
5. Give cos $(\pi/3)$.
6. Give the period of sin.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Review a half-turn on the circle; R85. **2.** Read the x-coordinate and y-coordinate of the point (1;0). **3.** Read the sign of the y-coordinate on the upper semicircle; R86. **4.** Locate the two points with x-coordinate zero. **5.** Use a equilateral triangle cut in half. **6.** Go round a full turn and compare two y-coordinates; R85. The R questions serve as review material with the teacher, even if their whole-class routine is scheduled later.

Objectives

Use parity and periodicity; differentiate sine and cosine; solve equations and inequalities; study a periodic function.

Course notes

Circle and functions

On the trigonometric circle, the point identified by an angle x in radians has coordinates $(\cos x, \sin x)$. For every real x,

$$\cos^2 x + \sin^2 x = 1, \quad -1 \leq \cos x, \sin x \leq 1.$$

Cosine is even: $\cos(-x) = \cos x$. Sine is odd: $\sin(-x) = -\sin x$. The two functions are 2π-periodic.

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Derivatives and variations

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

$$(\sin(ax+b))' = a\cos(ax+b), \quad (\cos(ax+b))' = -a\sin(ax+b).$$

On $[-\pi,\pi]$, cosine increases on $[-\pi,0]$ from -1 to 1, then decreases on $[0,\pi]$ from 1 to -1. Sine decreases on $[-\pi,-\frac{\pi}{2}]$, increases on $[-\frac{\pi}{2},\frac{\pi}{2}]$, then decreases on $[\frac{\pi}{2},\pi]$.

Justification. Read the sign of sine and cosine on the circle, then apply the link between derivative and variations. Periodicity extends the curves; it does not remove the need to study an interval of sufficient length.

Equations and inequalities

If $|a|>1$, $\cos x = a$ and $\sin x = a$ have no real solution. For a known angle θ,

$$\cos x = \cos\theta \iff x = \theta + 2k\pi \text{ or } x = -\theta + 2k\pi,$$

$$\sin x = \sin\theta \iff x = \theta + 2k\pi \text{ or } x = \pi - \theta + 2k\pi, \quad k\in\mathbb{Z}.$$

The solutions are then restricted to the required interval and duplicates are removed.

Examples on $[-\pi,\pi]$. $\cos x = \frac{1}{2}$ gives $x = \pm\frac{\pi}{3}$; $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. $\cos x \le \frac{1}{2}$ gives $[-\pi,-\frac{\pi}{3}]\cup[\frac{\pi}{3},\pi]$; $\sin x \ge \frac{1}{2}$ gives $[\frac{\pi}{6},\frac{5\pi}{6}]$.

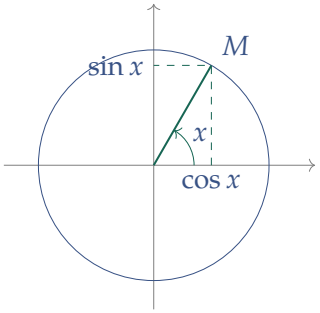
A trigonometric composite

To solve $\cos(2x) = \frac{1}{2}$ on $[-\pi,\pi]$, set $t = 2x \in [-2\pi,2\pi]$, solve for t, then divide by 2. We get $x \in \{-\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}\}$.

Optimisation. For $f(x) = \sin x + \frac{1}{2}\sin(2x)$ on $[0,\pi]$, $f' = \cos x + \cos(2x) = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1)$. The function increases up to $\frac{\pi}{3}$ then decreases; maximum $\frac{3\sqrt{3}}{4}$.

Double-angle identities used in the exercises. For every real x, $\sin(2x) = 2\sin x\cos x$ and $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$. These identities are recalled here so that their use in the problems is not left implicit.

Curve symmetries. The parity of cosine corresponds to a symmetry with respect to the y-axis; the oddness of sine to a point symmetry about the origin. Periodicity extends each curve by translation by the vector $(2\pi,0)$.



The coordinates of the point on the circle are $(\cos x, \sin x)$.

From the circle to the study of a function

5.1 One directed angle, two coordinates

First use: S75.

Definition · Sine and cosine

On the unit circle oriented anticlockwise, the point at angle x has coordinates $(\cos x; \sin x)$. Adding 2 $k\pi$ with $k \in \mathbb{Z}$ gives the same point. A negative measure corresponds to the opposite direction; it is not a negative length.

Worked example · Reading an equation and an inequality

On $[-\pi; \pi]$, $\cos x = 1/2$ corresponds to the two points with x-coordinate 1/2: $x = \pm\pi/3$. The inequality $\cos x \leq 1/2$ selects the arc to the left of the vertical line x = 1/2: $[-\pi; -\pi/3] \cup [\pi/3; \pi]$. The endpoints belong to the solution because the inequality is non-strict.

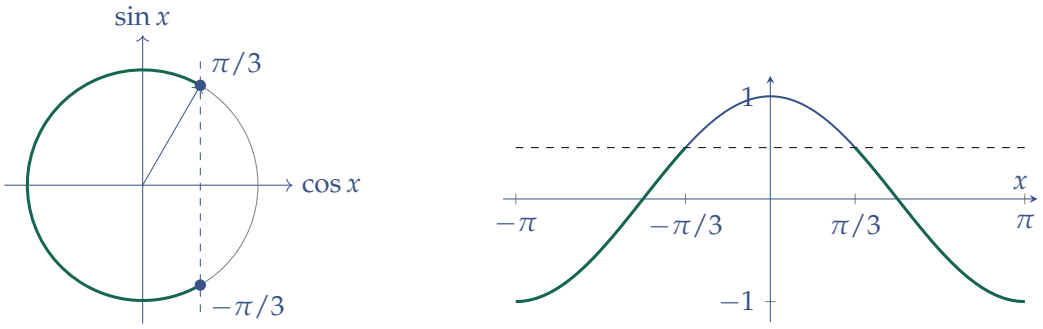


Figure 5.1. Same condition $\cos x \leq 1/2$: arc to the left of the vertical on the circle, curve below level 1 /2 to right.

A1 · Immediate application Solve $\sin x = 1/2$ on $[0; 2\pi[$, then $\sin x \geq 1/2$ on that same interval. Show the solutions on a circle and on a curve.

5.2 Reading amplitude and period

First use: S76.

Worked example · A shifted oscillation

For $h(t) = 3 + 2\sin(4t)$, the values lie between 1 and 5: the mean level is 3, the amplitude is 2. The smallest positive period is 2 $\pi/4 = \pi/2$. The number 2 π remains a period, but is not the smallest.

A2 · Partial guidance For $g(t) = 1 - 3\cos(2t)$, give the minimum, maximum, amplitude and smallest positive period. Find the values of t in $[0; \pi]$ where the extrema are attained.

A3 · Refute "The smallest positive period of $\sin^2 x$ is 2 π, like that of $\sin x$." Show that a translation by π leaves the function unchanged; explain why $\pi/2$ is not a period.

5.3 The sign comes from the circle

First use: S78.

Property · Derivatives and antiderivatives

The derivatives are $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$. For a composite, multiply by the inner derivative. An antiderivative of $\cos(ax+b)$ for $a \neq 0$ is $\sin(ax+b)/a$; an antiderivative of $\sin(ax+b)$ is $-\cos(ax+b)/a$.

Worked example · Variation on an interval

For $f(x) = x + \cos x$ on $[0;\pi]$, $f' = 1 - \sin x \geq 0$, with equality only at $\pi/2$. The function is strictly increasing: it strictly increases on each side of this isolated point and remains continuous. Moreover $f'' = -\cos x$ changes sign at $\pi/2$, so the curve has an inflection at $(\pi/2;\pi/2)$. A horizontal tangent is not necessarily an extremum.

A4 · Choice of strategy Study $f(x) = x - \sin x$ on $[0;\pi]$: derivative, variations, convexity. Deduce $\sin x \leq x$ on this interval. Does the formula $f'(0) = 0$ imply a change in variation?

5.4 An identity put to work on an area

First use: S79.

Worked example · Area of a rectangle

A point in the first quadrant of the unit circle has coordinates $(\cos\theta;\sin\theta)$, $0 \leq \theta \leq \pi/2$. The rectangle symmetric about the axes has area $A(\theta) = 4\sin\theta\cos\theta$. Since $(\sin\theta - \cos\theta)^2 \geq 0$ and $\sin^2\theta + \cos^2\theta = 1$, we have $2\sin\theta\cos\theta \leq 1$, therefore $A \leq 2$. Equality requires $\theta = \pi/4$: the optimal rectangle is a square.

A5 · Synthesis Determine by differentiation the maximum of $4\sin\theta\cos\theta$ on $[0;\pi/2]$. Recover the previous conclusion, then calculate $\int_0^{\pi/2} \sin\theta\cos\theta\,\mathrm{d}\theta$.

A6 · Independent problem Prove that $\cos x = x$ has a unique solution on $[0;\pi/2]$. Bound it between 0.7 and 0.8 using the calculator in radian mode. Suggest a method to improve the precision.

A first step: Study the difference of the two sides and separate existence, uniqueness and approximation.

One minute, spoken aloud. Explain how the same set of solutions appears as arcs on the circle and intervals on the x-axis of the curve.

Delayed review · S84 then S90. Mix limits, differentiation, equations, areas and trigonometry in a single study; redo a solution on another interval, checking the endpoints.

Investigation

Height of a point on a wheel - 35 min.

A wheel turns at constant speed in a vertical plane. Its radius is 2 m and its centre is 3 m above the ground. A point on the wheel is at its lowest position at time $t = 0$. The wheel completes one revolution in 8 s. Its height is modelled by $h(t) = 3 - 2\cos\left(\frac{\pi t}{4}\right)$, where t is expressed in seconds and $h(t)$ in metres.

1. Check that the model gives the stated initial height, a period of 8 s and the heights expected minimum and maximum.

2. Calculate $h'(t)$, then study the variations of h on $[0;8]$.

3. Determine all the times $t \in [0;8]$ for which the height is 4 m.

4. Determine the total time, during this revolution, for which the point is at least 4 m off ground.

Progressive exercises

E1 · Direct application. Solve in $[-\pi;\pi]$ the equation $\sin x = -\frac{1}{2}$.

E2 · Application and justification. Solve in $[-\pi;\pi]$ the inequality $\cos x \geq \frac{\sqrt{2}}{2}$.

E3 · Method to organise. We define on $\mathbb{R}$ the functions $f(x) = \cos(3x-1)$ and $g(x) = \sin^2 x$. Calculate their derivatives.

E4 · Choice of strategy. Consider $f(x) = 2 + \sin x$ on $[0;2\pi]$. Calculate its derivative and draw its table of variations.

E5 · Synthesis or proof. Calculate $\int_0^{\pi} \sin x \, \mathrm{d}x$, then the average value of the sine function on $[0;\pi]$.

Work from the textbook or a past paper

Barbazo, p. 166, Exercise 19. In S75: question 1, to read a symmetry and a period on a curve. At home: question 2, to prove the conjectures using identities valid for all reals.

Homework

DM5 - Splitting an area - 55 min, after S79.

Consider the function f defined on $[0;\frac{\pi}{2}]$ by $f(x) = \sin x \cos x$. Its graph is drawn in a orthonormal coordinate system.

1. Calculate $f'(x)$ and determine the maximum of f over this interval.

2. Solve on $[0;\frac{\pi}{2}]$ the inequality $f(x) \geq \frac{1}{4}$.

3. Calculate $\int_0^{\frac{\pi}{2}} f(x) \, \mathrm{d}x$, then the average value of f on $[0;\frac{\pi}{2}]$.

4. For $a \in [0;\frac{\pi}{2}]$, set $A(a) = \int_0^{a} f(x) \, \mathrm{d}x$. Determine a so that the region under the curve between 0 and a has an area equal to half the total area under the curve between 0 and $\frac{\pi}{2}$.

MCQ A · circle and functions

After S76; k denotes an integer.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $\cos(-x)$ is:

A. $-\cos x$ B. $\cos x$

C. $\sin x$ D. 0

2. The smallest strictly positive period of $x \mapsto \sin(3x)$ is:

A. 2π B. 3π

C. $\frac{2\pi}{3}$ D. $\frac{\pi}{3}$

3. The solutions of $\sin x = 0$ are:

A. $x = \frac{\pi}{2} + 2k\pi$ B. $x = k\pi$

C. $x = 2k\pi$ only D. $x = \pi/4$

4. The derivative of $\cos x$ is:

A. $\sin x$ B. $-\sin x$

C. $-\cos x$ D. $\tan x$

5. On $[0;\pi]$, the sine is:

A. always negative B. positive or zero

C. always increasing D. always zero

MCQ B · equations and function study

After S79.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. Solve $\cos x = 1$ on $[0;2\pi]$ gives:

A. 0 and 2π B. π

C. $\frac{\pi}{2}$ D. a single solution

7. To solve $\sin x \geq \frac{1}{2}$, in particular we need:

A. choose the correct arc, then intersect with the interval required B. divide by $\sin x$

C. use a binomial D. forget the endpoints

8. A function of the form $a + b\cos(\omega t)$, with $\omega \neq 0$ has amplitude:

A. a B. b

C. $|b|$ D. ω

9. If $f'(x) = \cos x$, f is decreasing on:

A. $]0;\frac{\pi}{2}[$ B. $]\frac{\pi}{2};\frac{3\pi}{2}[$

C. $]\frac{3\pi}{2};2\pi[$ D. $[-\frac{\pi}{2};\frac{\pi}{2}]$

10. Two solutions differing by 2π represent:

A. distinct angles but the same point on the circle B. A contradiction

C. opposite sines D. forbidden values

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $\sin^2 x + \cos^2 x = 1$ for every real x.
2. $\sin x = \frac{1}{2}$ has only one solution on $[0; 2\pi]$.
3. $\sin(a+b) = \sin a + \sin b$ for all real numbers a, b.

Review of prior work

I can read the circle, exploit parity and period, solve with an interval and study a function. **À**
Watch for: forgetting an endpoint solution; confusing a reading from a graph with a proof of an identity.